

3	(a)	(i)	$GPE_{final} = -\frac{GM_{earth}(80)}{6.4 \times 10^6 + (1.5 \times 10^3) \times 10^3}$ Total energy on Earth's surface = total energy at LEO $KE_i + GPE_i + \text{energy supplied} = KE_f + GPE_f$ $\left(-\frac{GM_{earth}(80)}{6.4 \times 10^6}\right) + 3.0 \times 10^9 = KE_f + \left(-\frac{GM_{earth}(80)}{6.4 \times 10^6 + (1.5 \times 10^3) \times 10^3}\right)$ $KE_f = 3.0 \times 10^9 + (6.67 \times 10^{-11})(6.0 \times 10^{24})(80)\left(\frac{1}{7.9 \times 10^6} - \frac{1}{6.4 \times 10^6}\right)$ $KE_f = 2.1 \times 10^9 \text{ J}$	C1 C1 A1
		(ii)	<u>Centripetal force is provided by gravitational force</u> $F_c = F_G$ $F_c = \frac{GMm}{r^2}$ $F_c = \frac{6.67 \times 10^{-11}(6.0 \times 10^{24})(80)}{(7.9 \times 10^6)^2}$ $F_c = 510 \text{ N}$ (1m for both working and answer.)	B1 B1
	(b)	(i)	Gravitational potential at a point is the <u>work done per unit mass by external agent</u> in bringing a small test mass <u>from infinity to that point</u>.	B1

		(ii)	Since $F = -m \frac{d\phi}{dr}$, the negative of the gradient of the ϕ-r graph gives the force. <u>Positive gradient means gravitational force (vector) is negative</u> <u>Since gravitational force is negative when displacement is positive, it is in the opposite direction as the displacement from Earth</u> and hence is attractive. OR <u>Potential decreases nearer to the Earth.</u> <u>Gravitational force must always be in direction of decreasing potential</u> and points towards Earth and is attractive. OR <u>The gravitational potential is always negative, so the external force points in the opposite direction to displacement from infinity to that point / points away from the earth.</u> <u>Hence, the gravitational force points opposite to the external force towards the earth</u> and is an attractive one.	M1 A1 (M1) (A1) (M1) (A1)
		(ii)	$KE_{initial} + U_{initial} = KE_{final} + U_{final}$ $\frac{1}{2}mu^2 + m\phi_{initial} = \frac{1}{2}mv^2 + m\phi_{final}$ $\frac{1}{2}u^2 + (-6.25 \times 10^7) = 0 + 0$ $u = 11.2 \text{ km s}^{-1}$	M1 A1